Letters

RESEARCH LETTER

Insufficient Sleep Among US Adolescents Across Behavioral Risk Groups

Insufficient sleep among US adolescents is a major public health problem. Although studies documented recent declines in sleep duration across demographic groups,[1] it is unknown whether sleep loss has disproportionately affected adolescents with behavioral health risks associated with insufficient sleep, including electronic media use, mental health symptoms, substance use, sedentary activity, or victimization.[2] This study examined national trends in insufficient sleep from 2007 to 2023 and whether trends differed by demographic or behavioral risk subgroups.

Editor's Note

Supplemental content

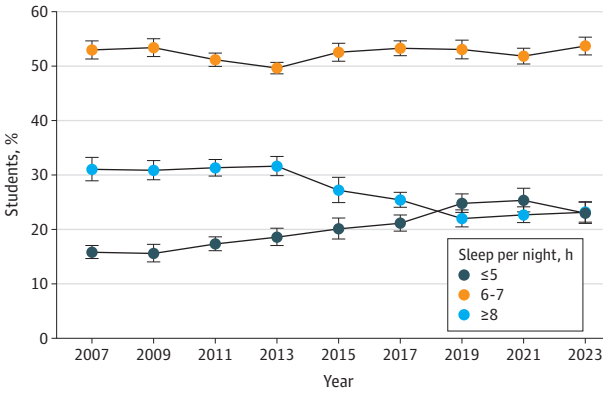
Methods | We analyzed cross-sectional data from the 2007-2023 national Youth Risk Behavior Survey, a biennial, school-based survey, to produce representative estimates of US high school students. Informed consent was obtained from the student's parent/guardian. The Centers for Disease Control and Prevention Institutional Review Board approved the Youth Risk Behavior Survey protocol. The study followed the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) reporting guideline.

The analytic sample included 120 950 students who answered a question about sleep duration on an average school night. Insufficient sleep was defined as less than or equal to 7 hours per night based on American Academy of Sleep Medicine guidelines[3]; very short sleep was defined as less than or equal to 5 hours per night. Self-reported demographic characteristics were included given known demographic variation in sleep duration.[4] Fifteen health risk behaviors associated with insufficient sleep were grouped into the following 5 categories: mental health, victimization, physical/sedentary, weight-related behaviors, and substance use. Additional details are shown in Supplement 1.

Trends in insufficient sleep, stratified by demographic and behavioral risk subgroups, were estimated using mean annual percent change (APC), adjusting for grade, sex, and race and ethnicity, using a modified Poisson model. Interactions between risk factors and time assessed whether trends differed by demographic and behavioral subgroups. Analyses were 2 sided, and P <.05 was considered significant. Analyses used Stata version 19.1 (StataCorp).

Results | The percentage of students reporting insufficient sleep increased from 68.9% in 2007 to 76.8% in 2023 (mean APC, 0.7% [95% CI, 0.6%-0.8%]). This increase was driven by a rise in very short sleep (≤5 hours), which increased from 15.8% to 23.0% (mean APC, 3.1% [95% CI, 2.5%-3.6%]), while the percentage reporting 6 to 7 hours of sleep did not change (**Figure**).

Figure. Sleep Patterns Among High School Students in the US



Data are from the Youth Risk Behavior Survey. Analyses were adjusted for complex survey designs. Bars represent 95% CIs.

Insufficient sleep (≤7 hours) increased across all demographic subgroups by sex, grade, and race and ethnicity (**Table**). Increases were significantly larger among non-Hispanic Black students than non-Hispanic White students. Similarly, insufficient sleep increased across nearly all behavioral subgroups, occurring as much or more among adolescents without behavioral risk factors as among those with them. Although insufficient sleep was consistently more common among adolescents reporting behavioral risks, the rate of increase was similar or greater among students without these risks. A few behaviors showed differential trends: insufficient sleep increased more among students reporting depressive symptoms or suicidal thoughts and increased less among students who watched television daily, used social media or video games at least 4 hours per day, or used alcohol or cigarettes (Table).

Discussion | These findings show a broad increase in insufficient sleep across all demographic groups, driven largely by increasing reports of very short sleep (≤5 hours). Notably, insufficient sleep increased as much or more among students without behavioral risks, suggesting that structural and environmental factors affecting most adolescents, rather than specific behaviors such as electronic media use, substance use, or sedentary activity, are contributing to widespread sleep loss. These trends highlight the need for population-level rather than targeted interventions. Later school start times, for example, have been associated with longer sleep, greater academic engagement, and improved mental health.[5] Despite the public health significance of declining adolescent sleep, intervention research remains limited.

This study has limitations, including that multiple hypotheses were tested without α adjustment; P values should be interpreted with caution. Other limitations include the inability to assess some potential behavioral contributors,

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Table. National Trends of Insufficient Sleep (≤7 Hours) Among High School Students in the US, 2007-2023

	Unweighted sample	Insufficient sleep, %		Mean APC (95% CI), %[a]	*P* value
		2007	2023		
Unweighted sample, No. (weighted %)		12 073 (9.4)	17 309 (14.0)		
Total	120 950	68.9	76.8	0.7 (0.6 to 0.8)	<.001
Grade					
Freshman	31 130	57.7	69.0	0.9 (0.7 to 1.1)	<.001
Sophomore	30 452	67.5	78.0	0.8 (0.6 to 0.9)	<.001
Junior	30 269	75.1	79.3	0.6 (0.4 to 0.7)	<.001
Senior	28 660	78.2	81.8	0.5 (0.3 to 0.6)	<.001
Female	60 398	71.3	78.3	0.6 (0.5 to 0.8)	<.001
Male	60 122	66.6	75.4	0.8 (0.6 to 0.9)	<.001
Sexual orientation		2015[b]			
Heterosexual	54 318	71.8	75.1	0.6 (0.3 to 0.8)	<.001
Gay or lesbian	1904	79.3	82.6	0.3 (−0.6 to 1.1)	.51
Bisexual	6234	76.2	83.8	0.7 (0.2 to 1.1)	.003
Not sure	4746	79.3	81.0	0.4 (−0.3 to 1.1)	.23
Race and ethnicity[c]					
Non-Hispanic American Indian or Alaska Native	2234	54.3	64.2	1.3 (0.2 to 2.4)	.03
Non-Hispanic Asian American, Native Hawaiian, or Other Pacific Islander	5802	74.1	80.3	0.6 (0.3 to 0.9)	<.001
Non-Hispanic Black	18 190	71.1	80.7	0.9 (0.6 to 1.1)	<.001[d]
Hispanic	30 284	65.5	76.3	0.8 (0.6 to 1.0)	<.001
Non-Hispanic White	56 371	69.2	75.5	0.6 (0.5 to 0.8)	<.001
Multiple races	6178	73.5	83.1	0.9 (0.5 to 1.2)	<.001
Depressive symptoms					
No	80 004	66.1	71.4	0.6 (0.4 to 0.7)	<.001
Yes	39 951	76.2	84.8	0.6 (0.5 to 0.8)	<.001[d]
Suicidal thoughts					
No	98 359	67.2	74.5	0.7 (0.5 to 0.8)	<.001
Yes	21 390	79.2	87.3	0.6 (0.5 to 0.8)	<.001[d]
Threatened with a weapon in the past year					
Never	111 150	68.4	76.3	0.7 (0.6 to 0.8)	<.001
1-3 Times	6269	75.9	84.0	0.8 (0.5 to 1.1)	<.001
≥4 Times	2483	72.9	79.5	0.4 (−0.0 to 0.9)	.10
Missed school due to safety issues in the past month					
Never	111 534	68.8	76.3	0.7 (0.6 to 0.8)	<.001
1-3 Times	7151	70.6	80.9	0.7 (0.4 to 1.0)	<.001
≥4 Times	1952	69.5	78.8	0.3 (−0.2 to 0.8)	.210
Bullied on a school property in the past year		2009[b]			
No	87 454	68.1	75.7	0.8 (0.7 to 0.9)	<.001
Yes	20 279	73.2	81.9	0.8 (0.6 to 1.0)	<.001
Bullied electronically in the past year		2011[b]			
No	78 854	67.2	75.1	0.9 (0.7 to 1.0)	<.001
Yes	14 353	75.4	84.9	0.8 (0.5 to 1.0)	<.001
Watching television on a daily basis			2019[b]		
Not at all	14 546	67.1	81.1	1.1 (0.8 to 1.4)	<.001
≤1 h	28 743	71.1	77.2	0.5 (0.3 to 0.8)	<.001[d]
2-3 h	30 809	67.1	77.2	0.7 (0.4 to 0.9)	<.001[d]
≥4 h	14 696	69.7	76.3	0.7 (0.4 to 1.0)	<.001[d]
Social media and video game use on a daily basis					
Not at all	17 313	65.6	69.2	0.8 (0.5 to 1.1)	<.001
≤1 h	31 081	67.2	74.7	0.6 (0.4 to 0.8)	<.001
2-3 h	36 043	69.9	78.2	0.5 (0.4 to 0.7)	<.001
≥4 h	31 747	76.1	78.6	0.4 (0.3 to 0.6)	<.001[d]

(continued)

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Table. National Trends of Insufficient Sleep (≤7 Hours) Among High School Students in the US, 2007-2023 (continued)

	Unweighted sample	Insufficient sleep, %		Mean APC (95% CI), %[a]	*P* value
		2007	2023		
Physical activity on a weekly basis					
Not at all	21 454	71.3	81.9	0.7 (0.6 to 0.9)	<.001
1-2 d	21 033	70.0	80.5	0.7 (0.6 to 0.9)	<.001
3-4 d	24 786	70.1	79.0	0.8 (0.6 to 0.9)	<.001
≥5 d	52 757	65.9	73.1	0.8 (0.6 to 0.9)	<.001
Soda consumption on a weekly basis					
Never	26 579	69.6	75.7	0.7 (0.5 to 0.9)	<.001
1-3 Times	39 151	67.4	76.4	0.9 (0.7 to 1.1)	<.001
4-6 Times	18 720	68.1	79.0	0.9 (0.7 to 1.1)	<.001
Every day	26 465	70.4	77.9	0.7 (0.5 to 0.8)	<.001
Underweight or normal (BMI <85th percentile)	76 215	69.2	76.9	0.7 (0.6 to 0.8)	<.001
Overweight (BMI 85th percentile to <95th percentile)	17 719	67.9	78.4	0.9 (0.7 to 1.0)	<.001
Obese (BMI ≥95th percentile)	16 788	70.0	80.5	0.8 (0.6 to 1.0)	<.001
Alcohol use in the past month					
Never	75 124	63.5	75.2	0.9 (0.8 to 1.1)	<.001
<10 d	32 346	74.4	81.2	0.6 (0.5 to 0.8)	<.001
≥10 d	4069	81.3	78.7	0.1 (−0.3 to 0.4)	.67[d]
Cigarette use in the past month					
Never	103 290	66.4	76.3	0.8 (0.7 to 1.0)	<.001
<10 d	7641	75.7	85.3	0.7 (0.4 to 1.0)	<.001
≥10 d	5506	80.3	83.9	0.0 (−0.4 to 0.3)	.93[d]
Cannabis use in the past month					
Never	94 587	66.9	75.1	0.8 (0.6 to 0.9)	<.001
<10 Times	13 930	78.1	83.9	0.6 (0.4 to 0.8)	<.001
≥10 Times	10 578	74.9	86.1	0.7 (0.4 to 0.9)	<.001

Abbreviation: BMI, body mass index (calculated as weight in kilograms divided by height in meters squared).

[a] Mean annual percentage change (APC) was estimated from biennial Youth Risk Behavior Survey data collected from 2007 through 2023 and adjusted for grade, sex, and race and ethnicity.

[b] Data from 2007 or 2023 were not available.

[c] Race and ethnicity were self-reported based on fixed categories.

[d] An interaction effect was observed when controlling for grade, sex, and race and ethnicity.

such as extracurricular demands and family conflict, the cross-sectional design that precludes causality or directionality of associations, and reliance on single-item measures of behavioral risks and self-reported sleep duration rather than actigraphy. In addition, because adolescents tend to overestimate sleep, true levels of insufficient sleep may be even higher.[6]

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